

CS 2300 HW 2

Problem 1

List the strong entity types in the ER diagram.

- BANK
- ACCOUNT
- LOAN
- CUSTOMER

Is there a weak entity type? If so, give its name, partial key, and identifying relationship. If so, specify the context (meaning) that the partial key and the identifying relationship of the weak entity type convey in this diagram.

- **Name:** BANK_BRANCH
- **Partial Key:** Branch_no
- **Identifying Relationship:** BRANCHES

Context and Meaning:

- The partial key Branch_no represents the branch number that is unique only within a specific bank, not across all banks in the system.
- The identifying relationship BRANCHES connects BANK_BRANCH to BANK, establishing that each branch must belong to exactly one bank and cannot exist independently.

List the names of all relationship types, and specify the (min, max) constraint on each participation of an entity type in a relationship type. Justify your choices.

BRANCHES

- **BANK participation:**
 - **(1, N):** A bank must have at least one branch (min=1) and can have many branches (max=N).

- **Justification:** A bank entity wouldn't exist without at least one branch. It can grow to have an unlimited number of branches.
- **BANK_BRANCH participation:**
 - **(1, 1):** A bank branch must belong to exactly one bank (min=1, max=1).
 - **Justification:** This is an identifying relationship for the weak entity BANK_BRANCH. A branch's existence is dependent on one specific bank, and it cannot be shared. The double line from BANK_BRANCH indicates total participation.

ACCTS

- **BANK_BRANCH participation:**
 - **(0, N):** A bank branch can have zero or many accounts (min=0, max=N).
 - **Justification:** The single line from BANK_BRANCH indicates partial participation, meaning a new branch might exist without having any accounts yet. A single branch can hold many different accounts.
- **ACCOUNT participation:**
 - **(1, 1):** An account must belong to exactly one bank branch (min=1, max=1).
 - **Justification:** The double line indicates total participation. Every account must be registered at a specific branch and cannot belong to more than one branch.

LOANS

- **BANK_BRANCH participation:**
 - **(0, N):** A bank branch can have zero or many loans (min=0, max=N).
 - **Justification:** The single line from BANK_BRANCH denotes partial participation; a branch is not required to have issued any loans. A branch can be responsible for many loans.
- **LOAN participation:**
 - **(1, 1):** A loan must be issued by and belong to exactly one bank branch (min=1, max=1).

- **Justification:** The double line signifies total participation, meaning every loan must be associated with a branch. A loan is a contract with a single branch.

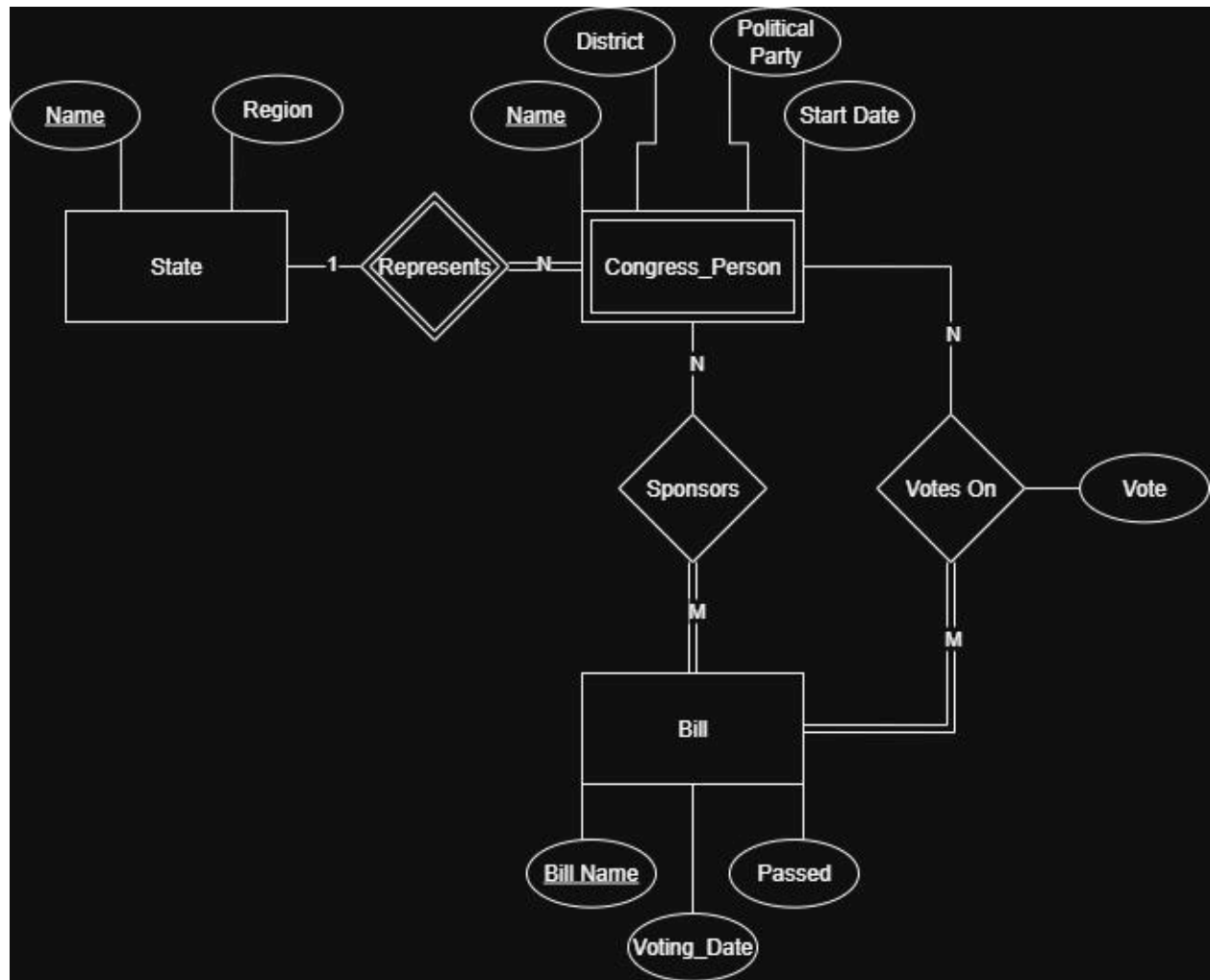
A_C

- **ACCOUNT participation:**
 - **(1, N):** An account must be owned by at least one customer (min=1) and can be jointly owned by many customers (max=N).
 - **Justification:** The double line indicates total participation; an account cannot exist without an owner. The 'N' on the CUSTOMER side of the M:N relationship allows for joint accounts.
- **CUSTOMER participation:**
 - **(1, M):** A customer must have at least one account (min=1) and can have many accounts (max=M).
 - **Justification:** The double line indicates that to be a customer of the bank in this context, one must hold an account. The 'M' on the ACCOUNT side allows a customer to have multiple accounts (e.g., checking, savings).

L_C

- **LOAN participation:**
 - **(1, N):** A loan must be associated with at least one customer (min=1) and can have multiple customers (co-borrowers) (max=N).
 - **Justification:** The double line signifies total participation; a loan cannot exist without a borrower. The 'N' on the CUSTOMER side allows for co-signers or joint borrowers on a single loan.
- **CUSTOMER participation:**
 - **(0, M):** A customer can have zero or many loans (min=0, max=M).
 - **Justification:** The single line from CUSTOMER indicates partial participation. A customer of the bank (who holds an account) is not required to have a loan. A customer is, however, permitted to take out multiple loans.

Problem 2



Key Assumptions:

Assumptions about Entities and Keys

- **State Uniqueness:** It is assumed that the Name of a state is unique and can serve as the primary key for the **STATE** entity.
- **Bill Uniqueness:** It is assumed that the Bill Name is unique for each bill and can act as its primary key.
- **Congress_Person is a Weak Entity:** The double-lined rectangle for **CONGRESS_PERSON** and the double-lined diamond for the **Represents**

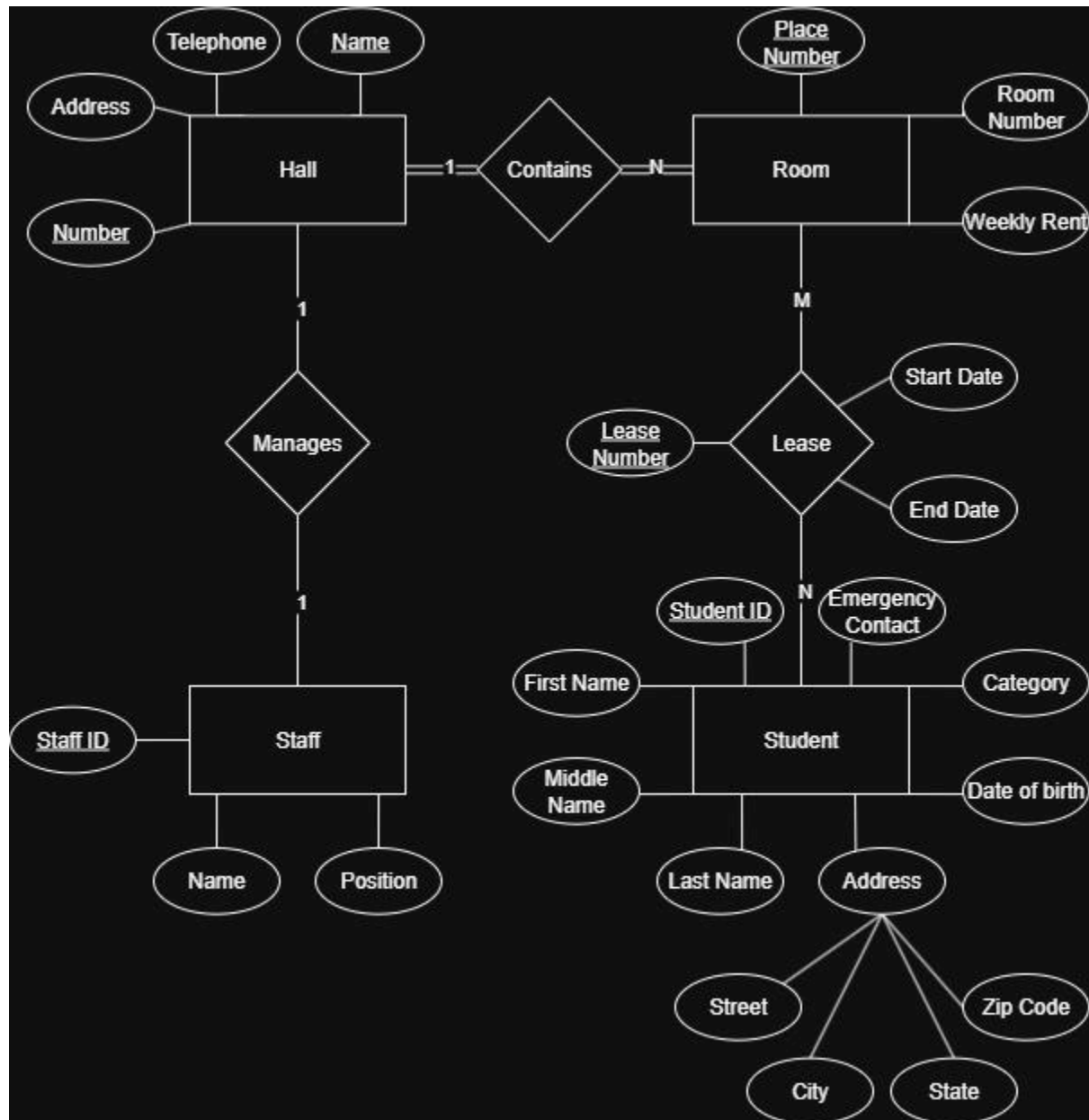
relationship signify a crucial assumption: A congressperson cannot be uniquely identified on their own. Their identity is dependent on the state they represent.

- **District as a Partial Key:** The District is assumed to be a partial key. It can uniquely identify a congressperson *only when combined with a specific state*. For example, "District 7" is not unique, but "California's District 7" is.

Assumptions about Relationships and Business Rules

- **Total Participation of a Bill in Relationships:**
 - The double line connecting the **BILL** entity to the **Sponsors** relationship implies an assumption of **total participation**. This means a bill cannot exist in the database unless it is sponsored by at least one congressperson.
 - Similarly, the double line connecting **BILL** to the **Votes On** relationship assumes **total participation**. This means every bill in the database must have been voted on by Congress.
- **Partial Participation of a Congress Person:**
 - The single lines connecting **CONGRESS_PERSON** to the **Sponsors** and **Votes On** relationships imply **partial participation**. This means a congressperson is not required to have sponsored a bill, nor are they required to have a recorded vote for every single bill (allowing for absences).
- **Nature of the 'Represents' Relationship:**
 - This is an **identifying relationship** that links the weak entity **CONGRESS_PERSON** to its strong owner entity, **STATE**.
 - The 1:N cardinality assumes that a state can have many representatives, but a representative can only represent one state and district combination.
 - The participation of **CONGRESS_PERSON** in this relationship is **total** (as is required for a weak entity); a congressperson must be associated with a state.

Problem 3



Key Assumptions:

Assumptions about Entities and Keys

- **Manager as an Entity:** It is assumed that a hall manager is a Staff member with their own distinct attributes (Staff ID, Name,

Position), rather than just a simple text attribute of the Hall. This justifies the creation of a separate Staff entity.

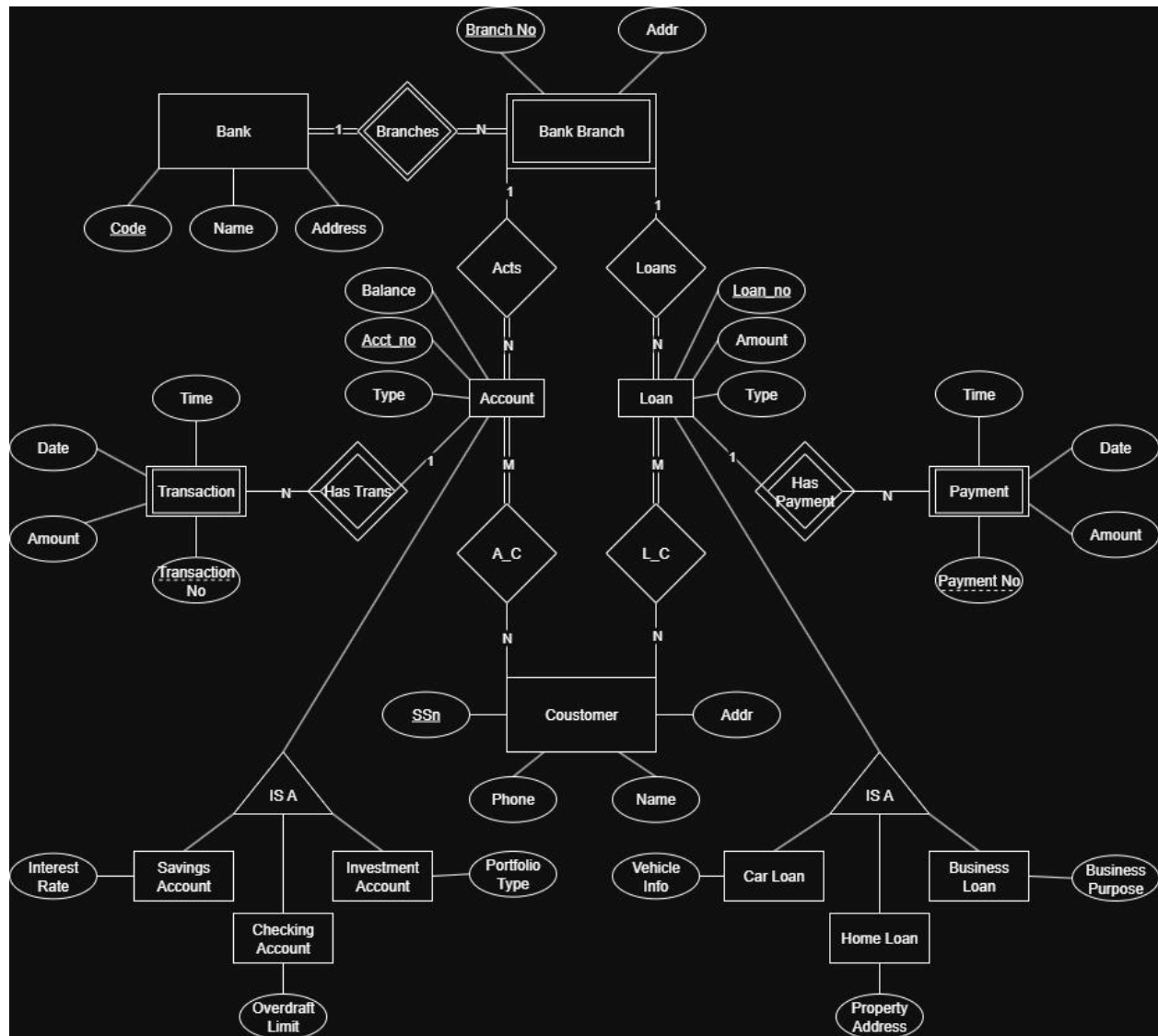
- **Unique Identifiers:** The diagram assumes that the following attributes are unique and can serve as primary keys for their respective entities:
 - Number for the Hall entity.
 - Staff ID for the Staff entity.
 - Place Number for the Room entity (explicitly stated in the requirements and underlined).
 - Student ID for the Student entity (underlined).
 - Lease Number for the Lease relationship.
- **Composite and Multi-valued Attributes:**
 - It's assumed that storing the components of a Student's Address (Street, City, State, Zip Code) separately is a requirement.
 - It is assumed that a Student can have more than one Emergency Contact, which is why it is modeled as a multi-valued attribute (double oval).

Assumptions about Relationships and Business Rules

- **Manages Relationship:**
 - It's assumed that every Hall must be managed by **exactly one** Staff member (total participation from Hall).
 - It's assumed that a Staff member can manage **at most one** Hall, and not all staff members are managers (partial participation from Staff).
- **Contains Relationship:**
 - The double lines indicate an assumption of **total participation** on both sides. This means:
 - A Hall must contain at least one Room to be considered a hall.
 - A Room must belong to exactly one Hall.
- **Lease Relationship:**
 - The M:N structure between Room and Student assumes that a room can be rented by many different students over time, and a student can rent different rooms over their university career (e.g., a new room each year).

- The attributes on the Lease relationship (Lease Number, Start Date, End Date) assume that the rental contract itself is a central concept that needs to be tracked over time.
- The single lines from Room and Student to the Lease relationship imply **partial participation**. This means:
 - A Room can be vacant and is not required to be part of a lease.
 - A Student is not required to live in university housing and may not have a lease.

Problem 4



Key Assumptions:

Of course. This is an excellent, well-formed EER diagram. Based on its structure and the specific notations used, here is a clear list of the assumptions made in this design.

Assumptions about Generalization/Specialization

- **Total and Disjoint Specialization:** The diagram assumes that the specializations for both ACCOUNT and LOAN are **total and disjoint**.
 - **Total:** This means every ACCOUNT entity *must* be one of the specified subtypes (either a Savings Account, Checking Account, or Investment Account). The same rule applies to LOANs (every loan must be a Car Loan, Home Loan, or Business Loan). An account or loan cannot exist without being one of these specific types.
 - **Disjoint:** This means that an entity can belong to **only one** of the subclasses. For example, a single account cannot be both a Savings Account and a Checking Account simultaneously.
- **Inheritance:** It is assumed that the subclasses inherit all the attributes and relationships of their respective superclasses.

Assumptions about Weak Entities

- **Existence Dependency:** The double-lined rectangles for Transaction and Payment and the double-lined diamonds for their relationships signify an assumption: **Transaction and Payment are weak entities**.
 - This means a Transaction cannot exist in the database without being associated with a specific ACCOUNT.
 - Similarly, a Payment cannot exist without being associated with a specific LOAN.
- **Partial Keys:** The dashed underlines for Transaction No and Payment No indicate that they are **partial keys**.
 - This assumes that Transaction No is only unique *per account*. For example, both account 101 and account 102 could have a "Transaction No. 5". The globally unique identifier for a transaction is the combination of the Acct_no and the Transaction No. The same logic applies to Payment No.

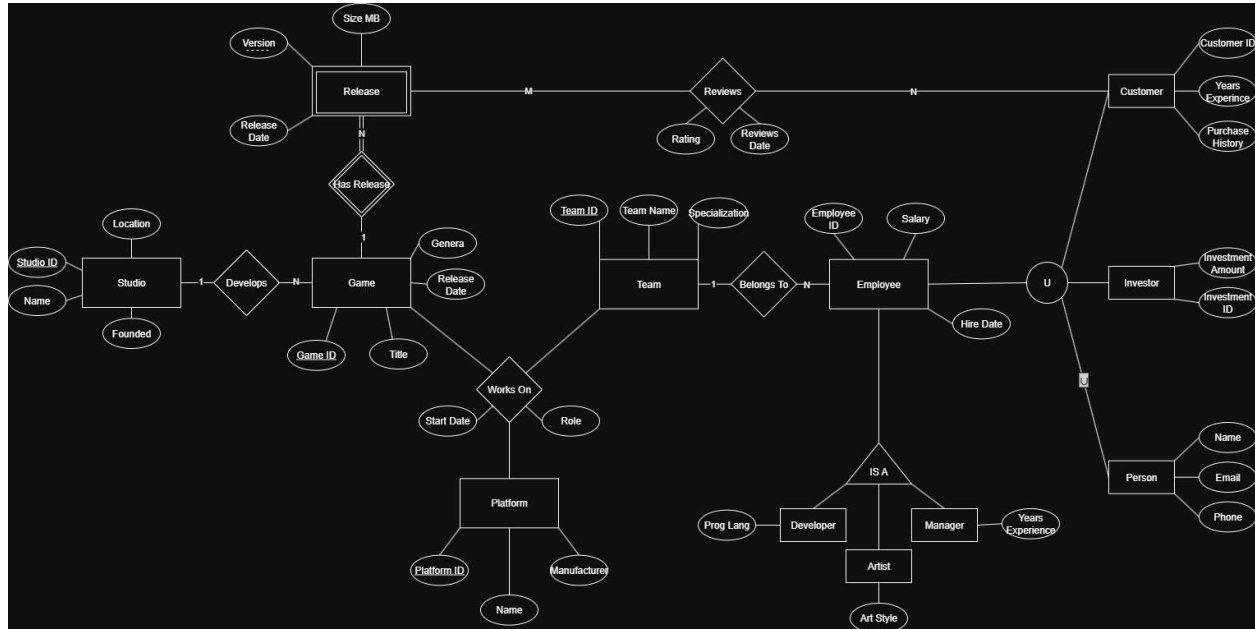
Assumptions about Relationships and Participation

- **Bank Branch as a Weak Entity:** The diagram maintains the original assumption that Bank Branch is a weak entity whose identity is dependent on the Bank it belongs to.

- **Partial Participation of Superclasses:** The single lines from ACCOUNT to Has Trans and from LOAN to Has Payment imply **partial participation**.
 - This assumes that a new ACCOUNT can be created before it has any transactions.
 - This assumes that a new LOAN can be created before any payments have been made against it.

Problem 5

Video Game Development Studio Database



Entity Types:

1. **STUDIO**: Game development companies
2. **GAME**: Video games being developed
3. **TEAM**: Development teams within studios
4. **PLATFORM**: Gaming platforms
5. **EMPLOYEE**: Studio staff (superclass)
6. **RELEASE**: Specific game releases/versions (weak entity)
7. **PERSON**: Union
8. **CUSTOMER**: Game purchasers
9. **INVESTOR**: Studio investors

Relationship Types:

1. **DEVELOPS** (1:N): Studio develops Games
2. **BELONGS_TO** (N:1): Employees belong to Teams
3. **HAS_RELEASE** (1:N): Games have Releases (identifying relationship)
4. **REVIEWS** (M:N): Customers review Games
5. **WORKS_ON** (Ternary): Teams work on Games for Platforms

Specialized Features:

Weak Entity Type:

- **RELEASE** depends on **GAME** via **HAS_RELEASE** identifying relationship
- Partial key: Version
- Complete key: Game_ID + Version

Superclass/Subclass Relationship:

- **EMPLOYEE** (superclass) with ISA relationship
- Subclasses: **DEVELOPER**, **ARTIST**, **MANAGER**
- Constraints: Disjoint, Partial (some employees may not fit these categories)

Category (Union):

- **PERSON** is a union of:
 - **EMPLOYEE**
 - **CUSTOMER**
 - **INVESTOR**
- Constraint: Total (every person must be at least one of these types)

N-ary Relationship:

- **WORKS_ON** is a ternary relationship connecting:
 - **TEAM**
 - **GAME**
 - **PLATFORM**
- Attributes: Start_Date, Role

Key Constraints:

Structural Constraints:

1. **Studio-Game** (1:N): Each studio develops multiple games
2. **Team-Employee** (1:N): Each team has multiple employees, each employee belongs to one team
3. **Game-Release** (1:N): Each game can have multiple releases

4. **Customer-Game Reviews:** Many-to-many with rating and review_date attributes

Participation Constraints:

- **Total:** RELEASE must participate in HAS_RELEASE
- **Total:** Every PERSON must be an EMPLOYEE, CUSTOMER, or INVESTOR
- **Partial:** EMPLOYEE participation in specialization (**Partial:** CUSTOMER participation in REVIEWS)
- **Partial:** GAME participation in REVIEWS

Key Constraints:

- **STUDIO:** Studio_ID (primary key)
- **GAME:** Game_ID (primary key)
- **TEAM:** Team_ID (primary key)
- **PLATFORM:** Platform_ID (primary key)
- **EMPLOYEE:** Employee_ID (primary key)
- **RELEASE:** Game_ID + Version (composite primary key)
- **CUSTOMER:** Customer_ID (primary key)
- **INVESTOR:** Investor_ID (primary key)
- **PERSON:** Shared attributes (Name, Email, Phone) across all categories

Additional Functional Dependencies:

- Game_ID -> Studio_ID (each game is developed by exactly one studio)
- Employee_ID -> Team_ID (each employee belongs to exactly one team)
- Team_ID -> Studio_ID (each team belongs to exactly one studio, derived through DEVELOPS and BELONGS_TO relationships)